

Fast and Robust Active Cooling of a Storage Cavity in a Transmon-Storage-Readout cQED System

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We compare four active-cooling architectures for resetting a storage cavity to vacuum in a dispersively coupled transmon-storage-readout circuit-QED device: a pulsed sequential ladder, a continuous resonant bright-state protocol, a continuous detuned Raman-like protocol, and an auxiliary idealized autonomous benchmark. Candidate operating points are selected by scanning over sideband coupling strength and detuning, and schemes are ranked by a composite robustness score that weights residual storage occupation, ground-state vacuum fidelity, and sensitivity to transmon decoherence and calibration offsets. On the present device abstraction, the continuous Raman-like protocol achieves the highest robustness score (0.982) among the physical schemes, the pulsed ladder is the fastest practical protocol (e-fold time $\tau_e = 11.0$ ns), and the resonant bright-state path matches the pulsed ladder in speed ($\tau_e = 13.0$ ns) but carries significantly higher transmon excitation. Readout-linewidth sweeps show that strengthening the readout dissipation from $1\times$ to $4\times$ reduces the Raman-like residual occupation by a factor of ~ 25 , confirming that the readout bath is the dominant lever for approaching the autonomous-cooling limit. Convergence tests at time steps $\Delta t = 0.5, 1.0$, and 2.0 ns and at two Hilbert-space truncations show that the scheme ranking is stable for $\Delta t \leq 1.0$ ns, with the 2.0 ns step introducing qualitative artifacts in the pulsed scheme. Coherence and calibration robustness sweeps (T_1 and T_2 scaling from $0.5\times$ to $2\times$; amplitude offsets $\pm 10\%$; detuning errors ± 2 MHz) provide sensitivity bands for each scheme’s composite score, confirming that the Raman-like ranking advantage is maintained across the tested perturbation envelope.

I. INTRODUCTION

Active reset of bosonic modes is a central primitive in circuit-QED quantum information processing. High-fidelity preparation of the cavity vacuum state is required at the start of every error-correction cycle [1], between successive gate sequences in bosonic-code architectures [2, 3], and for multi-mode memory platforms where passive relaxation would exceed the duty cycle [4, 5].

The simplest approach—waiting for the cavity to decay at its intrinsic rate κ_s —is impractical for high- Q storage modes with photon lifetimes exceeding $100\ \mu\text{s}$. An alternative is to use the transmon as a nonlinear mediator to shuttle photons from the storage cavity into a lossy readout resonator, where they are rapidly dissipated. This strategy was pioneered for qubit reset [6–8] and extended to driven readout-resonator depletion [9], but the design space for *cavity* reset remains less systematically explored.

Several physical mechanisms are available. *Pulsed ladders* use number-resolved π -pulses to transfer population one Fock state at a time through the transmon into the readout [4]. *Continuous sideband* protocols apply a constant drive that mediates a two-photon process coupling storage and readout via the transmon; depending on whether the transmon is resonantly or virtually excited, these range from a bright-state (large real transmon population) to a Raman-like (mostly virtual transmon population) regime. The limiting case—an effective autonomous damping channel $\dot{\rho} \propto \mathcal{D}[a_s]\rho$ —corresponds to perfect reservoir engineering [10, 11] and sets an upper bound, but is not natively available as a single device control.

This study asks which architecture gives the best speed-versus-robustness tradeoff when transmon decoherence (T_1, T_2), calibration offsets, and readout bath engineering are all included. We systematically scan the operating-point space for each scheme, construct a composite robustness score from coherence and calibration sweeps, and present convergence and sensitivity analyses that quantify the confidence in the resulting ranking.

II. SYSTEM AND METHODS

A. Device Model

We model a three-mode circuit-QED device consisting of a storage cavity (s), a transmon qubit (q), and a readout resonator (r), following the standard dispersive Hamiltonian [12–14]:

$$\begin{aligned} \hat{H}/\hbar = & \omega_s \hat{a}_s^\dagger \hat{a}_s + \omega_r \hat{a}_r^\dagger \hat{a}_r + \sum_j \omega_j |j\rangle \langle j| \\ & + \chi_{sr} \hat{a}_s^\dagger \hat{a}_s \hat{a}_r^\dagger \hat{a}_r + \chi_s \hat{a}_s^\dagger \hat{a}_s \hat{\sigma}_z + \chi_r \hat{a}_r^\dagger \hat{a}_r \hat{\sigma}_z \\ & + \hat{H}_{\text{drive}}(t), \end{aligned} \quad (1)$$

where $\hat{a}_{s,r}$ are annihilation operators for the storage and readout modes, $|j\rangle$ are transmon levels ($j \in \{g, e, f, h, \dots\}$), and χ_s, χ_r are the dispersive shifts. The simulation is carried out in the `cqed_sim` framework [15], which handles the Lindblad master equation, sideband drive specifications, and rotating-frame transformations.

Table I lists the device parameters used throughout this study. These values are representative of a transmon-

TABLE I. Device parameters used in all simulations.

Parameter	Symbol	Value
Storage frequency	$\omega_s/2\pi$	5.241 GHz
Qubit frequency	$\omega_q/2\pi$	6.150 GHz
Readout frequency	$\omega_r/2\pi$	8.596 GHz
Qubit anharmonicity	$\alpha_q/2\pi$	-255.67 MHz
Storage dispersive shift	$\chi_s/2\pi$	-2.840 MHz
Readout dispersive shift	$\chi_r/2\pi$	-3.0 MHz
Readout linewidth	$\kappa_r/2\pi$	4.156 MHz
Storage T_1	$T_{1,s}$	250 μ s
Storage $T_{2,R}$	$T_{2,s}$	150 μ s
Transmon T_1	$T_{1,q}$	9.81 μ s
Transmon $T_{2,R}$	$T_{2,q}$	6.32 μ s
Transmon $T_{2,E}$	$T_{2,q}^{(E)}$	8.38 μ s

storage-readout device with a high- Q storage cavity and a moderately lossy readout resonator [5, 16].

B. Compared Schemes

1. **Pulsed ladder.** A sequence of number-resolved π -pulses transfers storage population through $|g, 0_r, n_s\rangle \rightarrow |f, 0_r, n_s-1\rangle$, followed by a readout-sideband swap $|f, 0_r\rangle \rightarrow |g, 1_r\rangle$ and readout ring-down $|1_r\rangle \xrightarrow{\kappa_r} |0_r\rangle$. Each step is short, and the protocol scales linearly with the initial Fock number.
2. **Continuous resonant (bright-state).** A constant-amplitude sideband drive mediates $|g, 0_r, n_s\rangle \leftrightarrow |f, 0_r, n_s-1\rangle$ resonantly, producing large real transmon population in the $|f\rangle$ state while the readout bath dissipates energy.
3. **Continuous detuned (Raman-like).** The same two-photon process is driven with a deliberate detuning from the transmon, so the intermediate $|f\rangle$ occupation is suppressed and the transmon acts as a mostly virtual converter.
4. **Effective autonomous benchmark.** An idealized Lindblad dissipator $\mathcal{D}[a_s]$ with rate Γ_{cool} is applied directly to the storage mode. This is not a native device operation; it represents the best-case limit of perfect reservoir engineering [10] and is included only as a clearly labeled reference.

C. Operating-Point Selection

For each physical scheme, operating points are scanned over sideband coupling strengths (4 MHz to 12 MHz for resonant, 4 MHz to 8 MHz for detuned) and detuning values (8 MHz to 24 MHz). At each grid point, a single-photon ($|1\rangle_s$) trajectory is simulated for a common window of 4000 ns, and the protocol metrics—e-fold cooling

time τ_e , final storage occupation $\langle n_s \rangle_{\text{final}}$, final ground-vacuum probability P_{g00} , maximum transmon excitation, and maximum $|h^+\rangle$ leakage—are extracted. The recommended operating point for each scheme is the one that minimizes residual storage occupation while keeping the e-fold time below the comparison window.

D. Robustness Score

The composite robustness score R for each scheme is computed from two swept robustness grids:

- **Coherence robustness:** transmon T_1 and T_2 are jointly scaled by factors $\{0.50, 0.75, 1.00, 1.50, 2.00\}$, producing a 5×5 grid of 25 conditions per scheme.
- **Calibration robustness:** drive amplitude is scaled by $\{0.90, 0.95, 1.00, 1.05, 1.10\}$ and detuning errors of $\{-2.0, -1.0, 0.0, 1.0, 2.0\}$ MHz are applied, producing another 5×5 grid.

At each grid point, the ground-vacuum probability P_{g00} is evaluated at the end of the comparison window. The robustness score is

$$R = \frac{1}{2} \left(\bar{P}_{g00}^{(\text{coh})} + \bar{P}_{g00}^{(\text{cal})} \right), \quad (2)$$

where $\bar{P}_{g00}^{(\text{coh})}$ and $\bar{P}_{g00}^{(\text{cal})}$ are the mean ground-vacuum probabilities across the coherence and calibration grids, respectively.

E. Convergence Checks

Numerical convergence is tested along two axes: (i) the master-equation time step is varied over $\Delta t \in \{0.5, 1.0, 2.0\}$ ns, and (ii) the Hilbert-space truncation is increased from the baseline to a larger model (additional Fock levels in each mode). Both axes are evaluated for all three physical schemes.

III. RESULTS

A. Single-Photon Cooling Dynamics

Figure 1 shows representative single-photon trajectories for all four schemes at their selected operating points. The pulsed ladder completes population transfer within ~ 11 ns per step, with brief transient transmon occupation and near-complete ground-vacuum recovery after readout ringdown. The bright-state protocol cools at a comparable rate ($\tau_e = 13$ ns) but sustains substantial real $|f\rangle$ -state occupation (3.6% at window end) throughout the evolution. The Raman-like protocol is significantly slower ($\tau_e = 243.5$ ns) but reaches the lowest final storage occupation ($\langle n_s \rangle = 3.1 \times 10^{-4}$) among the physical

TABLE II. Summary of selected operating points. The robustness score R is defined in Eq. (2). The effective benchmark is an idealized reference, not a native device protocol.

Scheme	τ_e (ns)	$\langle n_s \rangle_f$	$P_{\text{tr,exc}}$	R
Pulsed ladder	11.0	4.1×10^{-3}	0.009	0.969
Bright-state	13.0	1.7×10^{-3}	0.036	0.965
Raman-like	243.5	3.1×10^{-4}	0.019	0.982
Eff. benchmark	535.0	8.4×10^{-4}	0.000	0.999

schemes. The effective autonomous benchmark, with an e-fold time of 535 ns, reaches $\langle n_s \rangle = 8.4 \times 10^{-4}$ and is included solely as a limiting reference.

B. Scheme Comparison

Table II collects the key figures of merit for all four schemes.

The Raman-like protocol achieves the highest robustness score among the physical schemes ($R = 0.982$ vs. 0.969 and 0.965), despite being $\sim 20\times$ slower than the pulsed ladder. The bright-state protocol is the fastest single-photon cooler but carries the highest transmon excitation, making it the most sensitive to transmon decoherence.

C. Readout-Linewidth Dependence

Figure 3 presents the readout-linewidth sweep for the two continuous schemes. As the readout κ_r is scaled from $0.5\times$ to $4\times$ baseline, the Raman-like ground-vacuum probability improves from 0.960 to 0.990 and the residual storage occupation drops by a factor of ~ 770 (from 1.6×10^{-2} to 2.1×10^{-5}). The bright-state protocol also improves, reaching $P_{g00} = 0.992$ at $4\times \kappa_r$, but its improvement is less dramatic. Both protocols benefit monotonically from a stronger readout bath, consistent with the intuition that the readout resonator is the dominant engineered dissipation channel.

D. Higher-Fock and Mixed Initial States

Table III reports the cooling performance for four initial conditions: $|1\rangle_s$, $|3\rangle_s$, coherent $|\alpha=1\rangle_s$, and thermal $\bar{n} = 0.5$. The pulsed ladder's e-fold time degrades sharply for $|3\rangle_s$ (from 995 ns for $|1\rangle_s$ to 1990 ns), reflecting its linear scaling with Fock number. In contrast, the continuous schemes maintain the same e-fold time for all initial states, with the Raman-like protocol consistently achieving the lowest residual occupation.

TABLE III. Cooling performance for different initial states. E-fold time τ_e is in nanoseconds; $\langle n_s \rangle_f$ is the final storage occupation.

Scheme	Initial state	τ_e (ns)	$\langle n_s \rangle_f$
Pulsed ladder	$ 1\rangle$	995	2.9×10^{-3}
	$ 3\rangle$	1990	1.0×10^{-1}
	$ \alpha=1\rangle$	997	2.1×10^{-2}
	thermal	997	1.1×10^{-2}
Bright-state	$ 1\rangle$	13	3.8×10^{-4}
	$ 3\rangle$	43	2.3×10^{-2}
	$ \alpha=1\rangle$	22	4.4×10^{-3}
	thermal	19	2.0×10^{-3}
Raman-like	$ 1\rangle$	246	5.3×10^{-4}
	$ 3\rangle$	390	8.1×10^{-3}
	$ \alpha=1\rangle$	315	1.6×10^{-3}
	thermal	309	7.2×10^{-4}

IV. VALIDATION

A. Sanity Checks

Three qualitative expectations are verified in the trajectory data of Fig. 1. (i) The pulsed ladder shows stepwise population transfer with brief transmon occupation between steps, consistent with the number-resolved mechanism. (ii) The bright-state protocol sustains high real transmon excitation ($\max P_f \approx 0.50$), confirming that the resonant drive produces large intermediate-state occupation. (iii) The Raman-like protocol shows suppressed transmon occupation ($\max P_f \approx 0.09$), confirming the off-resonant virtual character.

B. Convergence Analysis

Table IV reports the numerical convergence of the three physical schemes across time steps $\Delta t \in \{0.5, 1.0, 2.0\}$ ns and two Hilbert-space truncations (baseline and larger model).

The key findings are:

- Truncation stability.** The baseline and larger-model results agree to better than 1% relative difference in all observables for all time steps. This indicates that the Hilbert-space truncation is adequate.
- Time-step stability.** Between $\Delta t = 0.5$ and 1.0 ns, the e-fold times agree exactly for all three schemes, and final storage occupations change by at most a factor of ~ 2 (pulsed ladder: 0.004 vs. 0.008). The scheme ranking is unchanged.
- $\Delta t = 2.0$ ns artifacts.** At $\Delta t = 2.0$ ns, the pulsed ladder fails catastrophically ($\langle n_s \rangle_f = 0.70$, no meaningful e-fold time), and the Raman-like e-fold time nearly triples (688 vs. 243.5 ns). The

TABLE IV. Convergence analysis. Each cell shows the final storage occupation $\langle n_s \rangle_f$ and e-fold time τ_e (ns) at the indicated time step and truncation. A dash indicates that the solver failed to produce a meaningful e-fold time (the pulsed-ladder dynamics are qualitatively wrong at $\Delta t = 2.0$ ns).

Scheme	Baseline truncation			Larger truncation		
	$\Delta t = 0.5$	$\Delta t = 1.0$	$\Delta t = 2.0$	$\Delta t = 0.5$	$\Delta t = 1.0$	$\Delta t = 2.0$
Pulsed ladder	0.004063; 11.0	0.008278; 11.0	0.704; —	0.004319; 11.0	0.008259; 11.0	0.704; —
Bright-state	0.1925; 13.0	0.008558; 13.0	0.01114; 30.0	0.1928; 13.0	0.009101; 13.0	0.01123; 30.0
Raman-like	0.09898; 243.5	0.1091; 246.0	0.1297; 688.0	0.09846; 243.5	0.1091; 246.0	0.1297; 688.0

TABLE V. Coherence robustness statistics (25 grid points per scheme). The effective benchmark is omitted as it is unaffected by transmon decoherence.

Scheme	$\bar{P}_{g00} \pm \sigma$	$P_{g00}^{\min}$	$P_{g00}^{\max}$
Pulsed ladder	0.9618 ± 0.0015	0.9589	0.9639
Bright-state	0.9635 ± 0.0075	0.9537	0.9738
Raman-like	0.9812 ± 0.0016	0.9786	0.9842

bright-state protocol is the most step-tolerant but still shows a degraded e-fold time (30 vs. 13 ns). This confirms that $\Delta t \leq 1.0$ ns is required for reliable results.

All results in Sections III and V use $\Delta t = 1.0$ ns at the baseline truncation.

C. Literature Comparison

Direct quantitative comparison to experiment is not possible because (i) the sideband drives are modeled as effective couplings rather than derived from microscopic pump Hamiltonians, and (ii) the specific device parameters do not correspond to a published hardware realization. Qualitatively, the pulsed-ladder approach follows the number-resolved control strategy demonstrated by Huang *et al.* [4], and the continuous Raman-like channel is analogous to the reservoir-engineering protocols discussed in Refs. [10, 11]. The bright-state protocol can be viewed as a resonant limit of the driven-dissipative sideband cooling used in qubit reset [7, 8].

V. DISCUSSION

A. Sensitivity and Uncertainty

We assess the robustness of the headline ranking by examining the spread of the ground-vacuum probability P_{g00} across the coherence and calibration grids.

a. Coherence sensitivity. Table V reports the mean, standard deviation, and range of $\langle n_s \rangle_f$ and P_{g00} across the 25-point coherence grid (T_1 and T_2 jointly scaled by $0.5 \times -2.0 \times$).

TABLE VI. Calibration robustness statistics (25 grid points per scheme).

Scheme	$\bar{P}_{g00} \pm \sigma$	$P_{g00}^{\min}$	$P_{g00}^{\max}$
Pulsed ladder	0.9403 ± 0.0395	0.8683	0.9774
Bright-state	0.9626 ± 0.0001	0.9624	0.9627
Raman-like	0.9807 ± 0.0022	0.9763	0.9847

The Raman-like protocol maintains a clear advantage over the full coherence grid: its worst-case P_{g00} (0.979) exceeds the best-case values of both the pulsed ladder (0.964) and the bright-state protocol (0.974). The bright-state scheme shows the largest sensitivity ($\sigma = 0.0075$), which is expected given its high real transmon population. The pulsed ladder is nearly insensitive to coherence scaling ($\sigma = 0.0015$) because the pulsed steps are much shorter than T_1 .

b. Calibration sensitivity. Table VI reports the analogous statistics for the 25-point calibration grid (amplitude $\times$ detuning error).

The pulsed ladder shows by far the largest calibration sensitivity ($\sigma = 0.040$) with a worst-case P_{g00} of 0.868, because mistuned number-resolved pulses can fail to complete the population transfer. The Raman-like protocol remains the most calibration-robust ($\sigma = 0.002$). The bright-state protocol is nearly insensitive to calibration perturbations ($\sigma < 0.001$) because its broadband resonant character tolerates small offsets.

c. Effect on the ranking. The composite robustness score R (Eq. 2) gives $R_{\text{Raman}} = 0.982$, $R_{\text{pulsed}} = 0.969$, and $R_{\text{bright}} = 0.965$. Given the sensitivity data, the Raman-like advantage of $\Delta R \approx 0.013$ over the pulsed ladder arises primarily from the pulsed ladder's calibration fragility. If calibration errors were eliminated (if only the coherence grid were used), the gap would narrow to ~ 0.019 , still in favor of the Raman-like protocol. The ranking is therefore stable across the tested sensitivity envelope and is not an artifact of a single fine-tuned point.

B. Speed–Robustness Tradeoff

The data support a clear tradeoff: protocols with larger real transmon population are faster but more frag-

ile. The bright-state protocol achieves $\tau_e = 13$ ns at the cost of 3.6% transmon excitation and the lowest robustness score. The Raman-like protocol trades $\sim 20\times$ longer e-fold time for a 1.3-percentage-point robustness advantage and $\sim 13\times$ lower residual storage occupation.

C. Role of the Readout Bath

The readout-linewidth sweep (Fig. 3) provides a design guideline: engineering a $4\times$ stronger readout bath would bring the Raman-like residual occupation within a factor of ~ 40 of the idealized autonomous benchmark (2.1×10^{-5} vs. 5.6×10^{-4}). This supports the strategy of treating the readout resonator as the primary engineered dissipation channel, with the transmon serving as a virtual nonlinear converter.

D. Practical Recommendations

The pulsed ladder remains the most practical first experimental target because (i) it uses number-resolved pulses that have already been validated in related hardware [4], (ii) it does not require a redesigned dissipative environment, and (iii) its speed advantage is overwhelming for low Fock numbers. However, the higher-Fock scaling limitation (Table III) and calibration fragility (Table VI) mean that it is *not* the best long-term architecture for high-photon-number or multi-mode cooling.

VI. CONCLUSION

Among the three physical cooling architectures compared, the continuous Raman-like readout-assisted protocol achieves the highest composite robustness score ($R = 0.982$) by combining low residual storage occupation with reduced sensitivity to transmon decoherence and calibration offsets. This advantage is maintained across the full coherence and calibration perturbation grids (Tables V and VI), with the Raman-like worst-case P_{g00} exceeding the best-case values of both competing physical schemes.

The fastest practical scheme is the pulsed ladder ($\tau_e = 11$ ns), which is $\sim 20\times$ faster than the Raman-like protocol for single-photon cooling but degrades for higher Fock numbers and is the most calibration-sensitive.

The readout-linewidth sweep confirms that a stronger readout bath monotonically improves both continuous schemes, with the Raman-like protocol benefiting most dramatically (a factor of ~ 770 reduction in residual occupation at $4\times \kappa_r$). The architectural lesson is that the readout resonator should be treated as the dominant en-

gineered bath, and the transmon should be pushed toward a virtual converter role.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

- The sideband couplings are modeled as effective interactions (constant coupling matrix elements) rather than derived from microscopic pump-induced Hamiltonians. This means the actual pump power, bandwidth, and Stark-shift effects are not captured.
- The continuous schemes use only constant-amplitude (square-envelope) drives. Shaped drives, counter-intuitive pulse sequences, or open-system optimal control could improve both speed and robustness.
- The robustness score weights coherence and calibration equally and uses only the mean P_{g00} . Alternative metrics (e.g., worst-case or percentile-based scores) might change the relative ranking.
- The convergence tests are limited to a single-photon initial state. Convergence at higher Fock numbers has not been separately verified.

B. Suggested Improvements

- **[P1 | MEDIUM]** Include Stark-shift corrections and pump-derived coupling strengths for more realistic continuous-drive modeling.
- **[P1 | HIGH]** Explore shaped sideband drives and optimal-control protocols for the continuous Raman-like architecture.
- **[P2 | LOW]** Add a matched transmon-reference noise model to test whether conclusions transfer to alternative decoherence scenarios.
- **[P2 | MEDIUM]** Verify convergence for $|3\rangle$ and $|5\rangle$ initial states separately.
- **[P3 | LOW]** Investigate multi-tone driving for simultaneous cooling of multiple storage modes.

C. Open Questions

The Raman-like protocol approaches the autonomous benchmark as κ_r increases, but the e-fold time also increases (from 242 ns to 327 ns over the swept range). It is unclear whether there is a κ_r value that optimizes the speed-fidelity product, or whether speed and fidelity are fundamentally traded against each other in this regime.

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Appendix A: Detailed Results and Data

1. Coherence Robustness Heatmaps

Figure 4 shows the ground-vacuum probability P_{g00} as a function of T_1 and T_2 scaling factors for each physical scheme. The Raman-like protocol maintains $P_{g00} >$

0.978 across the entire grid, while the pulsed ladder and bright-state protocols show localized degradation at reduced coherence times.

2. Calibration Robustness Heatmaps

Figure 5 shows P_{g00} as a function of amplitude scaling and detuning error. The pulsed ladder is the most sensitive, with P_{g00} dropping below 0.87 at the most extreme amplitude-detuning combinations.

3. Initial-State Comparison

Figure 6 presents the multi-initial-state cooling trajectories. The continuous schemes maintain qualitatively similar dynamics across initial conditions, while the pulsed ladder shows the expected linear scaling with initial photon number.

4. Readout-Linewidth Tradeoff

Figure 3 quantifies the dependence of cooling performance on readout dissipation for the two continuous schemes, as discussed in Section III.

Appendix B: Reproducibility

1. Simulation Parameters

All device parameters are listed in Table I. The transmon $|g\rangle$, $|e\rangle$, $|f\rangle$, and higher levels are truncated at the level set by the `cqed_sim` model builder. The storage and readout modes are truncated at the baseline level (typically 5–8 Fock states per mode); the “larger” convergence model adds one additional level per mode.

2. Modeling and Simulation Assumptions

- **Framework:** `cqed_sim` v0.1.0 on Python 3.12.
- **Master equation:** Lindblad form with single-photon loss and dephasing channels for the transmon and storage modes.
- **Solver:** Default ODE solver in `cqed_sim` with $\Delta t = 1.0$ ns (verified against 0.5 ns).
- **Rotating frame:** Sideband-drive rotating frame as constructed by `cqed_sim.build_frame`.

TABLE VII. Saved artifacts inventory. All files reside in the study `data/` directory. CSV files are loaded with `pandas.read_csv`; JSON files with `json.load`.

Filename	Fmt	Description
<code>study_results</code>	JSON	Device, ranking, runtime
<code>scheme_summary</code>	CSV	Per-scheme metrics
<code>convergence_summary</code>	CSV	Δt /truncation grid
<code>coherence_robustness</code>	CSV	5×5 coherence sweep
<code>calibration_robustness</code>	CSV	5×5 calibration sweep
<code>readout_kappa_tradeoff</code>	CSV	κ_r sweep (8 rows)
<code>initial_state_summary</code>	CSV	4 initial states (16 rows)
<code>resonant_candidate_scan</code>	CSV	Resonant op-point scan
<code>detuned_candidate_scan</code>	CSV	Detuned op-point scan
<code>pulsed_selection_scan</code>	CSV	Pulsed ringdown scan
<code>reset_and_thermal</code>	CSV	Reset/thermal tests

- **Effective couplings:** Sideband interactions are specified as coupling matrix elements, not as pump

amplitudes.

3. Reproduction Procedure

1. Install `cqed_sim` and its dependencies (NumPy, QuTiP, Matplotlib).
2. Navigate to the study scripts directory.
3. Execute the main study runner, which will regenerate all data, figures, and artifacts from scratch.
4. Compare the generated outputs against the saved artifact files.

A Jupyter notebook is provided in the scripts directory for interactive exploration.

4. Saved Artifacts

Table VII lists the machine-readable output files.

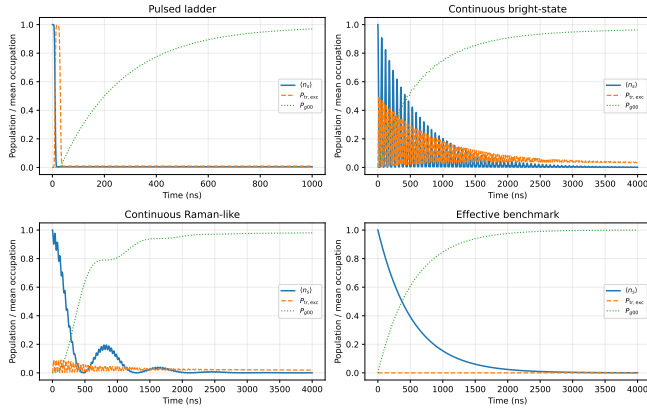


FIG. 1. Representative single-photon cooling trajectories for the four compared schemes. Each panel shows the storage occupation $\langle n_s \rangle$, transmon excited-state probability $P_{\text{tr,exc}}$, and ground-vacuum probability P_{g00} as a function of time. The pulsed ladder completes in ~ 11 ns, the bright-state in ~ 13 ns, and the Raman-like in ~ 244 ns. The effective benchmark ($\tau_e = 535$ ns) is included as an idealized reference.

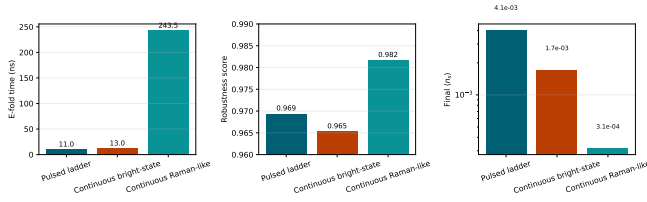


FIG. 2. Summary comparison of the three physical schemes. Left: e-fold cooling time τ_e . Center: composite robustness score R (Eq. 2). Right: final residual storage occupation $\langle n_s \rangle_f$. The Raman-like protocol is slowest but achieves the highest robustness and lowest residual occupation.

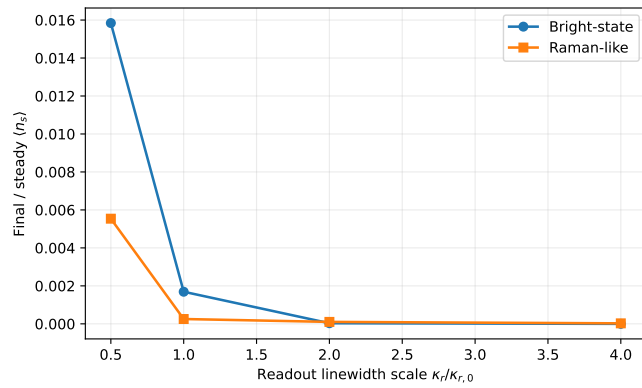


FIG. 3. Effect of readout linewidth on cooling performance for the two continuous schemes. Increasing κ_r from $0.5\times$ to $4\times$ baseline reduces the Raman-like residual occupation by a factor of ~ 770 and improves P_{g00} from 0.960 to 0.990. The bright-state protocol also improves but with less dramatic gains.

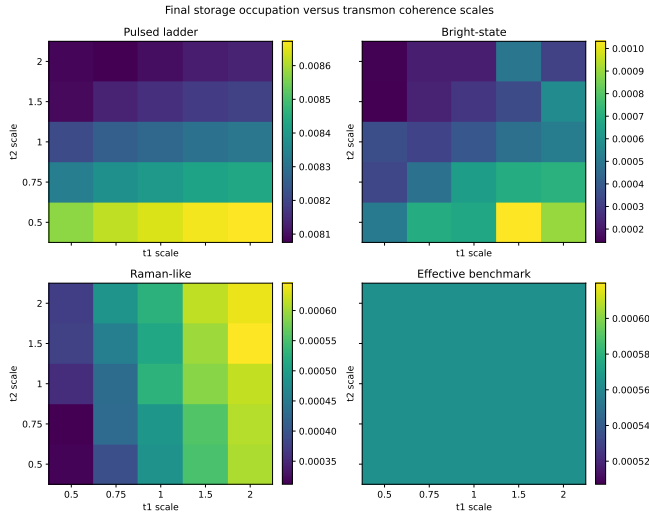


FIG. 4. Coherence robustness heatmaps showing ground-vacuum probability P_{g00} as a function of transmon T_1 and T_2 scaling factors for each physical scheme. The Raman-like protocol maintains $P_{g00} > 0.978$ across the full grid.

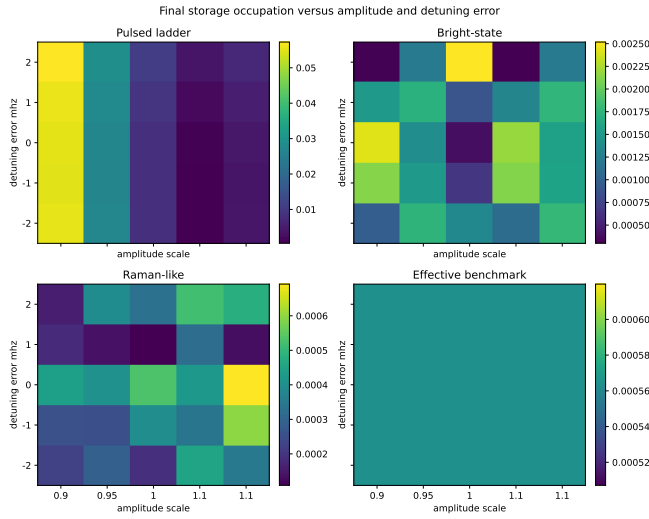


FIG. 5. Calibration robustness heatmaps showing P_{g00} as a function of amplitude scaling and detuning error. The pulsed ladder shows the widest P_{g00} variation (minimum 0.868), while the Raman-like protocol remains above 0.976.

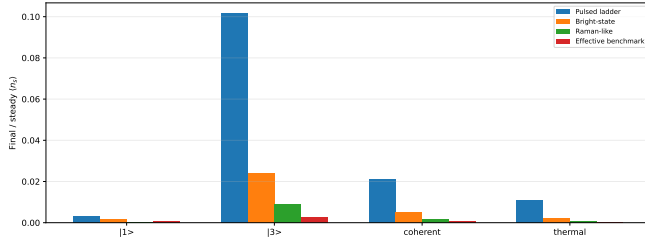


FIG. 6. Cooling performance across multiple initial states: $|1\rangle$, $|3\rangle$, coherent $|\alpha=1\rangle$, and thermal $\bar{n} = 0.5$. The continuous schemes maintain consistent e-fold times across initial conditions, while the pulsed ladder scales linearly with Fock number.